Nevil C Prince

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EDUCATION

Institution	Degree / Board	Passing Year	Score / CGPA
Indian Institute of Technology Hyderabad	B.Tech	2027	7.13
Rajagiri Christu Jayanthi Public School	Class XII (CBSE)	2022	92.40%
Rajagiri Christu Jayanthi Public School	Class X (CBSE)	2020	93.80%

WORK EXPERIENCE

Software Trainee Intern - Scadea Solutions

- Engineered **Synapse AI**, a **SharePoint-grounded** knowledge-access service for internal **knowledge-transfer** workflows, covering the full pipeline from **document ingestion** through **hybrid retrieval** to **answer generation**
- Built an **Answerability Verdict** module that detects unanswerable queries during the **answer pipeline**, cutting hallucinated or incorrect responses before they reach end users
- Designed a self-reinforcing **knowledge base** loop that captures resolved answers as **Captured Answers** and reindexes them into retrieval, shrinking recurring **Systemic Knowledge Gaps** over time

PROJECTS

Multi-Agent Stock Analysis and Investment Advisor | GitHub

- Designed a **six-agent equity-research pipeline** (technical, macro, fundamental, sentiment, risk, and portfolio) running as a **LangGraph DAG** behind a **FastAPI** gateway to generate research theses and allocations
- Built an **async data pipeline** that pulls **real-time market data** at configurable intervals and feeds **technical, macro, and sentiment signals** into the agent layer
- Used a **Gaussian HMM** to classify **macro regimes** and paired it with **deterministic risk models**, keeping allocation calls grounded in statistics instead of LLM guesswork
- Wired up **LangSmith tracing** across the LLM-driven agents, backed by **Pydantic validation** and an independent **kill switch/governor** to keep the system safe by default

Agentic RAG for SEC EDGAR Filings | GitHub

- Designed an **Agentic Corrective RAG (CRAG)** system in **LangGraph** to audit SEC 10-K filings, using a **two-stage relevance gate** and capped query rewrites to block hallucinated answers
- Built **hybrid retrieval** combining **Qdrant** dense vectors and **BM25S** sparse search, fused with **Reciprocal Rank Fusion** and reranked by a **BGE cross-encoder** for higher precision
- Orchestrated the **LangGraph** pipeline with **Qdrant semantic caching** and **hybrid model routing** - Groq LPUs for utility tasks, Gemini for final synthesis - cutting API costs by **85%**
- Implemented local **guardrails** with **Microsoft Presidio** PII detection and **Llama Prompt Guard 2** for prompt-injection defense, keeping sensitive filing data private and secure end-to-end

Business Entity Resolution | **Amazon ML Challenge 2026** | Github

- Developed a **blocking** stage for matching **1.7M businesses** against **10M records** across three countries, merging **word**, **character 3-gram**, and fine-tuned **E5 embedding** search with **Reciprocal Rank Fusion** to reach **99.6% candidate recall**
- Trained an **XGBoost** classifier on **26 name and address similarity features**, then added an **XLM-RoBERTa cross-encoder** and a **stacking model** to re-score each business's top candidates, improving held-out **F0.5** from **0.945 to 0.977**
- Implemented a decision step that selects the match set maximizing **expected F0.5** from calibrated probabilities, enforcing **one-to-one record assignment** and confirming generalization to an unseen country through **leave-one-country-out** testing

Optimization of UAV Swarm Operations for Persistent Surveillance | *Prof. Venkatraman Renganathan* | GitHub

- Formulated UAV fleet sizing as a convex **mixed-integer program**, minimizing weighted **launch-vs-spare** cost subject to an **inventory** equality constraint and a minimum **spare-floor ratio**
- Derived a linear **battery-endurance feasibility constraint** linking **ferry distance**, **patrol length**, **cruise velocity**, and a **state-of-charge floor** to guarantee single-flight energy feasibility

- Implemented a **Pyomo-based MILP solver** enforcing the inventory and spare-floor constraints, cross-validated against a **closed-form exact solution** to confirm optimality across fleet sizes
- Built a custom discrete-time-step **simulation engine** using **NumPy/Pandas** for route generation and analysis, sustaining 83-85% coverage with **sub-1-percentage-point** degradation after a single UAV failure

Quantum Gate Set Tomography With Graph Neural Networks | *Prof. Sangkha Borah* | GitHub

- Developed a **PyTorch**-based deep learning pipeline for **quantum gate set tomography**, replacing computationally intractable **Maximum Likelihood Estimation** with a scalable, fully differentiable neural approach
- **Modeled quantum circuits as directed graphs** using a **Relational Graph Convolutional Network (R-GCN)**, capturing both **temporal** gate sequencing and **spatial qubit entanglement**, benchmarked against a **Transformer** baseline
- Applied **Adaptive Layer Normalization (AdaLN)** to condition network layers on measured outcome probabilities, and used **curriculum learning** to progressively increase circuit depth for stable convergence
- Achieved precise prediction of **depolarizing error** and **over-rotation** parameters by backpropagating through a differentiable **Pauli Transfer Matrix** physics simulator, optimized via **KL divergence** and **MSE** losses

Aerodynamic Coefficient Prediction using ANN | *Prof. Lakshmana Dora Chandrala* | GitHub

- Generated and preprocessed **1.2M+** aerodynamic samples for **NACA airfoils** using **JavaFoil** simulations, then trained **Multi-Layer Perceptrons** in **TensorFlow/Keras** to predict **lift**, **drag**, and **pitching moment coefficients**
- Ran systematic experiments across varying **network depth and width** to identify the **optimal MLP architecture**, balancing prediction accuracy against training and inference cost
- Evaluated **regression model performance** across multiple **input resolutions**, achieving **$R^2 > 0.99$** for most predicted variables and confirming reliable accuracy on unseen **test data**
- Tested the model's ability to **generalize to an unseen data distribution**, achieving strong accuracy for most predicted variables, further improved through **combined-dataset training**

SKILLS

- **Programming Languages:** Python, SQL, C (Basics)
- **Backend & Databases:** FastAPI, Pydantic, AsyncIO, REST APIs, PostgreSQL (Supabase, pgvector), SQLite, Qdrant, ChromaDB
- **DevOps & AI:** Docker, Cloudflare, AWS, GitHub Actions (CI/CD), pytest, Linux, Git, LangGraph, TensorFlow, PyTorch, Hugging Face
- **CS Fundamentals:** DSA, OOP, DBMS, Operating Systems, System Design, Low-Level Design, Design Patterns

RELEVANT COURSES

Machine Learning (Stanford CS229), Artificial Intelligence, Multi-Agent Systems, Convex Optimization, Machine Learning for Physical Sciences, Data Science Analysis, Data Structures & Applications, Linear Algebra, Linear Systems & Signals, Probability & Random Variables, Computational Physics

POSITIONS OF RESPONSIBILITY

Powertrain Domain Head | **Aeolus Racing (Formula Student)** | **IIT Hyderabad**

- Led a **8-member** powertrain subteam to design, manufacture, and validate **SAE-compliant** subsystems
- Coordinated across vehicle domains to test physical components, optimize performance, and enforce track safety protocols

EXTRACURRICULAR

1st Place, Institute Water Polo League (2024): Won the championship tournament at IIT Hyderabad